

Pre-Algebra Practice Test II - Version One

Name
Date
Course

Instructions

- As always, write a title, name, date, and course.
- Number each problem, circle the number, and show all work.
- Draw a rectangle around your answers.
- Rewrite the equation whenever you perform an operation on both sides to preserve the result of the previous step.

Solve.

$$\textcircled{1} \quad -3 + x = 8$$

$$3 - 3 + x = 8 + 3$$

$$\boxed{x = 11}$$

$$\textcircled{2} \quad -12 = 4y$$

$$\frac{-12}{4} = \frac{4y}{4}$$

$$\boxed{-3 = y}$$

$$\textcircled{3} \quad z + 9 = 4$$

$$-9 + z + 9 = 4 - 9$$

$$\boxed{z = -5}$$

$$\textcircled{4} \quad -6 = \frac{x}{-8}$$

$$(-8)(-6) = \frac{x}{-8}(-8)$$

$$\boxed{48 = x}$$

$$\textcircled{5} \quad 2y - 3 = -6$$

$$3 + 2y - 3 = -6 + 3$$

$$2y = -3$$

$$\frac{2y}{2} = \frac{-3}{2}$$

$$\boxed{y = -\frac{3}{2}}$$

$$\textcircled{6} \quad -7 = 4 + \frac{z}{3}$$

$$-4 - 7 = 4 + \frac{z}{3} - 4$$

$$-11 = \frac{z}{3}$$

$$(3)(-11) = \frac{z}{3}(3)$$

$$\boxed{-33 = z}$$

$$\textcircled{7} \quad -10 - \frac{x}{4} = -5$$

$$10 - 10 - \frac{x}{4} = -5 + 10$$

$$\frac{x}{4} = 5$$

$$(-4)\frac{x}{4} = 5(-4)$$

$$\boxed{x = -20}$$

$$\textcircled{8} \quad 3 - 8y = 2$$

$$-3 + 3 - 8y = 2 - 3$$

$$-8y = -1$$

$$\frac{-8y}{-8} = \frac{-1}{-8}$$

$$\boxed{y = \frac{1}{8}}$$

$$\textcircled{9} \quad 5z - 8z + 7 = 16$$

$$-3z + 7 = 16$$

$$-7 - 3z + 7 = 16 - 7$$

$$-3z = 9$$

$$\frac{-3z}{-3} = \frac{9}{-3}$$

$$\boxed{z = -3}$$

$$\textcircled{10} \quad -1 = -6x + x + 1$$

$$-1 = -5x + 1$$

$$-1 - 1 = -5x + 1 - 1$$

$$-2 = -5x$$

$$\frac{-2}{-5} = \frac{-5x}{-5}$$

$$\boxed{\frac{2}{5} = x}$$

$$\textcircled{11} \quad -2y - 8y + 3 = 1$$

$$-10y + 3 = 1$$

$$-3 - 10y + 3 = 1 - 3$$

$$-10y = -2$$

$$\frac{-10y}{-10} = \frac{-2}{-10}$$

$$\boxed{y = \frac{1}{5}}$$

$$\textcircled{12} \quad 8 + z - 2z = 4$$

$$8 - z = 4$$

$$-8 + 8 - z = 4 - 8$$

$$-z = -4$$

$$\frac{-1z}{-1} = \frac{-4}{-1}$$

$$\boxed{z = 4}$$

$$\textcircled{13} \quad -3x + 5 + 2x - 8x = 7 - 8 + 3$$

$$-9x + 5 = 2$$

$$\textcolor{red}{-5} - 9x + 5 = 2 \textcolor{red}{-5}$$

$$-9x = -3$$

$$\frac{-9x}{\textcolor{teal}{-9}} = \frac{-3}{\textcolor{teal}{-9}}$$

$$\boxed{x = \frac{1}{3}}$$

$$\textcircled{14} \quad -1 + (-3) + 8 - 7 - 1 = 4 + 10y - 3 + 7y + (-y) + 1$$

$$-4 = 2 + 16y$$

$$\textcolor{red}{-2} - 4 = 2 + 16y \textcolor{red}{-2}$$

$$-6 = 16y$$

$$\frac{-6}{\textcolor{teal}{16}} = \frac{16y}{\textcolor{teal}{16}}$$

$$\boxed{-\frac{3}{8} = y}$$

$$\textcircled{15} \quad -7z + 4z + z - 3 = (-8) + 4 - 9$$

$$-2z - 3 = -13$$

$$\textcolor{red}{3} - 2z - 3 = -13 \textcolor{red}{+3}$$

$$-2z = -10$$

$$\frac{-2z}{\textcolor{teal}{-2}} = \frac{-10}{\textcolor{teal}{-2}}$$

$$\boxed{z = 5}$$

$$\textcircled{16} \quad -x + 8 = 1$$

$$\textcolor{red}{-8} - x + 8 = 1 \textcolor{red}{-8}$$

$$-x = -7$$

$$\frac{-1x}{\textcolor{teal}{-1}} = \frac{-7}{\textcolor{teal}{-1}}$$

$$\boxed{x = 7}$$

$$\textcircled{17} \quad -4z + 5z - 3 = -4$$

$$z - 3 = -4$$

$$\textcolor{red}{3} + z - 3 = -4 \textcolor{red}{+3}$$

$$\boxed{z = -1}$$

$$\textcircled{18} \quad \frac{z}{5} - 2 = -2$$

$$2 + \frac{z}{5} - 2 = -2 + 2$$

$$\frac{z}{5} = 0$$

$$(5)\frac{z}{5} = 0(5)$$

$$\boxed{z = 0}$$

$$\textcircled{19} \quad -4 = 10 - x$$

$$-10 - 4 = 10 - x - 10$$

$$-14 = -x$$

$$\frac{-14}{-1} = \frac{-1x}{-1}$$

$$\boxed{14 = x}$$

$$\textcircled{20} \quad \frac{3y}{4} + \frac{y}{8} - y = 4$$

$$\frac{6y}{8} + \frac{y}{8} - \frac{1y}{1} = 4$$

$$\frac{6y}{8} + \frac{y}{8} - \frac{8y}{8} = 4$$

$$-\frac{y}{8} = 4$$

$$(8)\left(-\frac{y}{8}\right) = 4(8)$$

$$-y = 32$$

$$\frac{-1y}{-1} = \frac{32}{-1}$$

$$\boxed{y = -32}$$

$$\textcircled{21} \quad -1 = \frac{2z}{3} - \frac{z}{4} + 3$$

$$-1 = \frac{8z}{12} - \frac{3z}{12} + 3$$

$$-1 = \frac{5z}{12} + 3$$

$$-3 - 1 = \frac{5z}{12} + 3 - 3$$

$$-4 = \frac{5z}{12}$$

$$(12)(-4) = \frac{5z}{12}(12)$$

$$-48 = 5z$$

$$\frac{-48}{5} = \frac{5z}{5}$$

$$\boxed{-48/5 = z}$$

$$\textcircled{22} \quad \frac{x}{6} + 1 - \frac{2x}{3} + \frac{5x}{9} + (-2) = -4 - 3 - 8$$

$$\frac{3x}{18} - \frac{12x}{18} + \frac{10x}{18} - 1 = -15$$

$$\frac{x}{18} - 1 = -15$$

$$1 + \frac{x}{18} - 1 = -15 + 1$$

$$\frac{x}{18} = -14$$

$$(18)\frac{x}{18} = -14(18)$$

$$\boxed{x = -252}$$

$$\begin{array}{r} 3 \\ 18 \\ \times 14 \\ \hline 172 \\ + 180 \\ \hline 252 \end{array}$$

$$\textcircled{23} \quad 5 - 7y = 5$$

$$-5 + 5 - 7y = 5 - 5$$

$$-7y = 0$$

$$\frac{-7y}{-7} = \frac{0}{-7}$$

$$\boxed{y = 0}$$

$$\textcircled{24} \quad -6 + 1 = -2z - 5z + 7$$

$$-5 = -7z + 7$$

$$-7 - 5 = -7z + 7 - 7$$

$$-12 = -7z$$

$$\frac{-12}{-7} = \frac{-7z}{-7}$$

$$\boxed{\frac{12}{7} = z}$$

$$\textcircled{25} \quad -6 - 8 - \frac{2x}{5} + \frac{x}{5} + \left(-\frac{x}{10}\right) + 4x = -7 + 1 - 8$$

$$-\frac{x}{5} + \left(-\frac{x}{10}\right) + \frac{4x}{1} - 14 = -14$$

$$-\frac{2x}{10} - \frac{x}{10} + \frac{40x}{10} - 14 = -14$$

$$\frac{37x}{10} - 14 = -14$$

$$14 + \frac{37x}{10} - 14 = -14 + 14$$

$$\frac{37x}{10} = 0$$

$$(10) \frac{37x}{10} = 0(10)$$

$$37x = 0$$

$$\frac{37x}{37} = \frac{0}{37}$$

$$\boxed{x = 0}$$